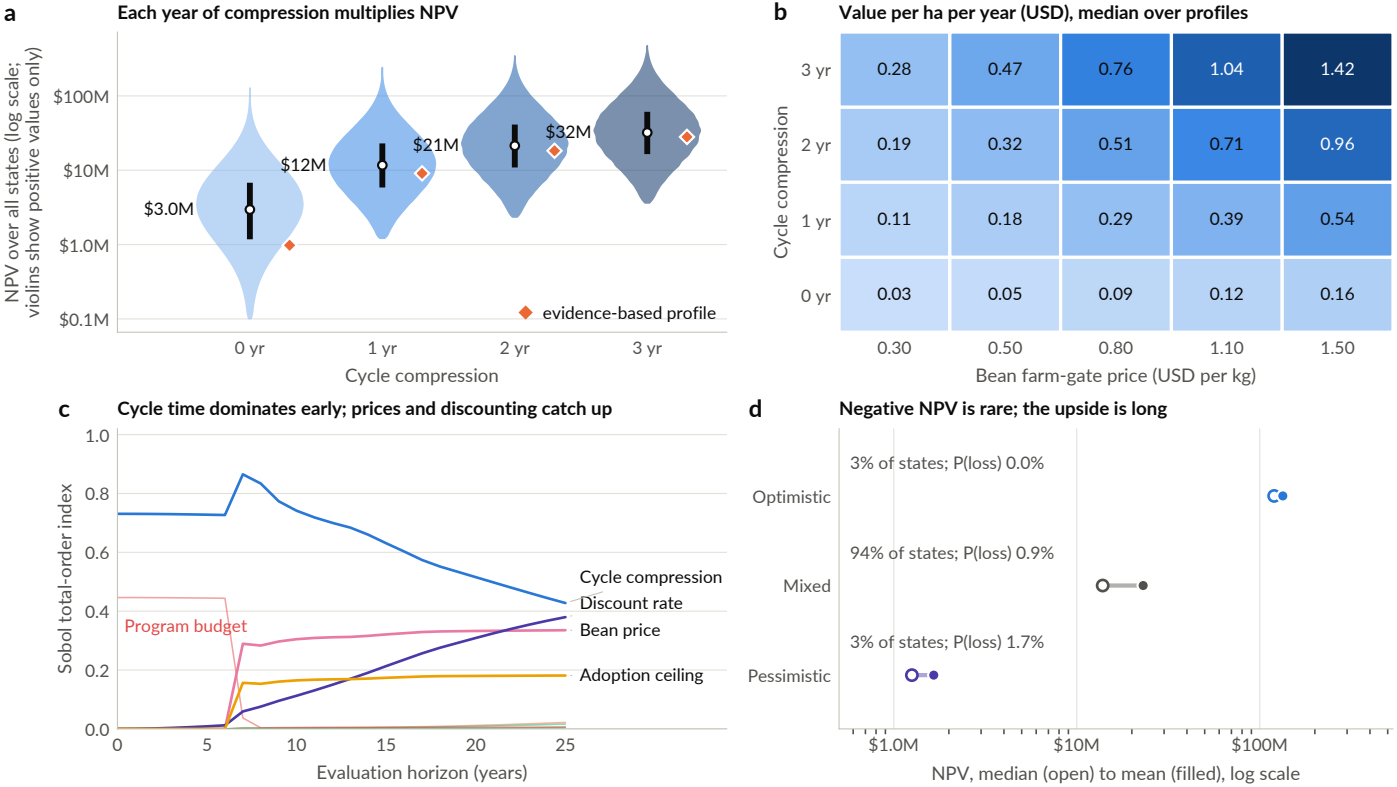
a
Each year of compression multiplies NPV
NPV over all states (log scale; violins show positive values only)
$100M
$10M
$1.0M
$0.1M
$3.0M
$12M
$21M
$32M
evidence-based profile
0 yr
1 yr
2 yr
3 yr
Cycle compression
b
Value per ha per year (USD), median over profiles
Cycle compression
3 yr
2 yr
1 yr
0 yr
0.28
0.47
0.76
1.04
1.42
0.19
0.32
0.51
0.71
0.96
0.11
0.18
0.29
0.39
0.54
0.03
0.05
0.09
0.12
0.16
0.30
0.50
0.80
1.10
1.50
Bean farm-gate price (USD per kg)
c
Cycle time dominates early; prices and discounting catch up
Sobol total-order index
1.0
0.8
0.6
0.4
0.2
0.0
Program budget
Cycle compression
Discount rate
Bean price
Adoption ceiling
0
5
10
15
20
25
Evaluation horizon (years)
d
Negative NPV is rare; the upside is long
3% of states; P(loss) 0.0%
Optimistic
94% of states; P(loss) 0.9%
Mixed
3% of states; P(loss) 1.7%
Pessimistic
$1.0M
$10M
$100M
NPV, median (open) to mean (filled), log scale